Product Requirements Document (PRD)

Cover Page

- **Project Name:** LayoverNow
  Course: CISC 593/594

- **Repository URL:** https://github.com/pranavapp2024-sys/LayoverNow.git
- **Current Branch:** main
- **Current Commit SHA: To Be Completed**
- **Current Release Version:** v2.0.0
- **Document Version:** 1.0
- **Last Updated:** 2026-07-21

Revision History

Version	Date	Git Commit	Description	Author
1.0	2026-07-21	**To Be Completed**	Initial generation of the PRD based on project README.	AI Assistant

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1. Product Vision

Problem Statement Standard flight search engines do not easily optimize for multi-day layovers, causing travelers to miss out on free, planned stopovers during flight connections.

Intended Users

- Leisure travelers
- Digital nomads
- Budget-conscious flyers

Stakeholders

- Project developers
- Travelers
- Partner Airlines with official stopover programs (Icelandair, TAP Portugal, Copa, Emirates)

Product Goals

- Surface optimal flight routes whose layover cities are worth stopping in.
- Provide interactive visualizations of candidate flight paths.
- Assist users with layover planning (visa, weather, itinerary) using AI context.

Major Features

- Trip Configuration (Airport autocomplete, date picker, stopover slider)

- Interactive Flight Radar
- Stopover Deals Ranking
- AeroAI Stopover Assistant

Planned Software Versions

- Version 1: Deterministic layover finder (offline)
- Version 2: AeroAI Advisor and live trip context

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2. Product Scope

Included functionality

- Offline airport database for origin/destination autocomplete.
- Haversine detour calculation and stopover hub scoring.
- Canvas-based interactive flight radar.
- Stopover Deals results ranked by Best Value, Savings, or Minimum Detour.
- AeroAI Stopover Assistant for visa requirements, currency, weather, and itineraries.
- Booking link generation for partner airlines.
- Graceful offline fallback mechanism.

Excluded functionality

- Hidden-city or skiplagging route calculations.
- Direct booking transactions within the extension.

Future enhancements

- **To Be Completed**

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3. Software Capabilities

3.1 Level-1 Capabilities

1 Configure Trip Parameters

2 Calculate Stopover Deals
3 Render Flight Radar
4 Generate Booking Links
5 Manage AI Assistant Context
6 Manage Fallback Mechanisms

3.2 Level-2 Capabilities

1 Configure Trip Parameters
    - 1.1 Autocomplete Airport Selection
    - 1.2 Select Departure Date
    - 1.3 Adjust Stopover Duration
2 Calculate Stopover Deals
    - 2.1 Calculate Haversine Detour
    - 2.2 Rank Stopover Hubs
3 Render Flight Radar
    - 3.1 Plot Candidate Flight Paths
    - 3.2 Update Active Coordinates
4 Generate Booking Links
    - 4.1 Construct Airline Booking URL
5 Manage AI Assistant Context
    - 5.1 Fetch Visa Requirements
    - 5.2 Fetch Weather Data
    - 5.3 Generate Day-by-Day Itinerary
6 Manage Fallback Mechanisms
    - 6.1 Detect API Failure
    - 6.2 Trigger Deterministic Fallback

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4. Undesirable Events

UE ID	Level-2 Capability	Undesirable Event
UE-1.1-01	1.1 Autocomplete Airport Selection	Autocomplete fails to find a valid airport code.
UE-1.2-01	1.2 Select Departure Date	User selects a past departure date.

UE ID	Level-2 Capability	Undesirable Event
UE-1.3-01	1.3 Adjust Stopover Duration	User sets stopover duration exceeding airline program limits.
UE-2.1-01	2.1 Calculate Haversine Detour	Haversine calculation fails on antimeridian crossing.
UE-2.2-01	2.2 Rank Stopover Hubs	Hub ranking returns no valid routes.
UE-3.1-01	3.1 Plot Candidate Flight Paths	Flight paths fail to render on the canvas.
UE-3.2-01	3.2 Update Active Coordinates	Active coordinates are stale or incorrect.
UE-4.1-01	4.1 Construct Airline Booking URL	Booking URL parameters are malformed.
UE-5.1-01	5.1 Fetch Visa Requirements	Visa API rate limit exceeded.
UE-5.2-01	5.2 Fetch Weather Data	Weather API returns missing data.
UE-5.3-01	5.3 Generate Day-by-Day Itinerary	AI hallucinates incorrect itinerary details.
UE-6.1-01	6.1 Detect API Failure	API failure detection times out too slowly.
UE-6.2-01	6.2 Trigger Deterministic Fallback	Deterministic fallback fails to activate.

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5. Risk Analysis

UE ID	Risk Statement	Likelihood	Impact	Risk Score
UE-1.1-01	If autocomplete fails to find a valid airport code, the user cannot initiate a search.	2	4	8
UE-1.2-01	If the user selects a past departure date, the search will yield no results or error out.	3	3	9
UE-1.3-01	If the stopover duration exceeds airline limits, generated booking links will fail.	3	4	12

UE ID	Risk Statement	Likelihood	Impact	Risk Score
UE-2.1-01	If Haversine calculation fails on antimeridian crossing, incorrect detours will be shown.	2	4	8
UE-2.2-01	If hub ranking returns no valid routes, the extension fails to provide its core value.	2	4	8
UE-3.1-01	If flight paths fail to render, the visual experience is degraded.	2	3	6
UE-3.2-01	If active coordinates are stale, the radar will display incorrect aircraft positions.	3	2	6
UE-4.1-01	If booking URL parameters are malformed, the user cannot complete their booking on the airline site.	2	5	10
UE-5.1-01	If the Visa API rate limit is exceeded, critical trip context is missing.	4	3	12
UE-5.2-01	If the Weather API returns missing data, the user misses environmental context.	3	2	6
UE-5.3-01	If AI hallucinates incorrect itinerary details, the user may rely on false planning information.	3	4	12

UE ID	Risk Statement	Likelihood	Impact	Risk Score
UE-6.1-01	If API failure detection times out too slowly, the UI will freeze and frustrate the user.	3	4	12
UE-6.2-01	If deterministic fallback fails to activate, the entire application becomes unusable during an outage.	2	5	10

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6. Risk Prioritization

Priority	UE ID	Risk Score
1	UE-1.3-01	12
2	UE-5.1-01	12
3	UE-5.3-01	12
4	UE-6.1-01	12
5	UE-4.1-01	10
6	UE-6.2-01	10
7	UE-1.2-01	9
8	UE-1.1-01	8
9	UE-2.1-01	8
10	UE-2.2-01	8
11	UE-3.1-01	6
12	UE-3.2-01	6
13	UE-5.2-01	6

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7. Risk Mitigation

UE ID	Risk Mitigation	Classification
UE-1.1-01	The Configuration Module shall constrain input to the offline airport database.	Pure Software

UE ID	Risk Mitigation	Classification
UE-1.2-01	The Configuration Module shall disable past dates in the date picker.	Pure Software
UE-1.3-01	The Configuration Module shall cap the stopover slider at 14 days.	Pure Software
UE-2.1-01	The Search Engine shall implement edge-case testing for antimeridian math.	Pure Software
UE-2.2-01	The UI shall display a friendly fallback message if no routes are found.	Pure Software
UE-3.1-01	The Radar Module shall catch canvas exceptions and fail gracefully.	Pure Software
UE-3.2-01	The Radar Module shall interpolate coordinates when active updates are delayed.	Pure Software
UE-4.1-01	The Booking Module shall validate URL parameters against known airline formats.	Pure Software
UE-5.1-01	The AI Context Module shall cache responses to minimize API calls.	Pure Software
UE-5.2-01	The UI shall hide the weather panel if data is unavailable.	Pure Software
UE-5.3-01	The AI Context Module shall prepend a disclaimer about AI-generated content.	Pure Software
UE-6.1-01	The Fallback Module shall enforce a strict 5-second timeout on all API requests.	Pure Software
UE-6.2-01	The application architecture shall use V1 offline capabilities as the default state.	Pure Software

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8. Functional Requirements

Requirement ID	Level-2 Capability	Functional Requirement
FR-1.1.1	1.1 Autocomplete Airport Selection	The Configuration Module shall process airport autocomplete queries within 100 milliseconds.

Requirement ID	Level-2 Capability	Functional Requirement
FR-1.2.1	1.2 Select Departure Date	The Configuration Module shall manage departure date selections within standard calendar constraints.
FR-1.3.1	1.3 Adjust Stopover Duration	The Configuration Module shall manage stopover slider bounds within 1 to 14 days.
FR-2.1.1	2.1 Calculate Haversine Detour	The Search Engine shall calculate Haversine detours within 1 second.
FR-2.2.1	2.2 Rank Stopover Hubs	The Search Engine shall perform stopover ranking within offline database limits.
FR-3.1.1	3.1 Plot Candidate Flight Paths	The Radar Module shall plot candidate flight paths within the canvas element.
FR-3.2.1	3.2 Update Active Coordinates	The Radar Module shall manage active coordinate updates within one radar sweep cycle.
FR-4.1.1	4.1 Construct Airline Booking URL	The Booking Module shall construct airline booking URLs within standard HTTP GET constraints.
FR-5.1.1	5.1 Fetch Visa Requirements	The AI Context Module shall fetch visa requirements within a 5-second timeout.
FR-5.2.1	5.2 Fetch Weather Data	The AI Context Module shall fetch weather data within a 5-second timeout.
FR-5.3.1	5.3 Generate Day-by-Day Itinerary	The AeroAI Assistant shall execute layover itinerary generation within API rate limits.
FR-6.1.1	6.1 Detect API Failure	The Fallback Module shall execute API failure detection within 5 seconds of the initial request.
FR-6.2.1	6.2 Trigger Deterministic Fallback	The Fallback Module shall manage the transition to deterministic ranking within 1 second of API failure.

9. Quality Requirements

- **Performance:** The extension popup shall render the initial UI in under 200 milliseconds.
- **Reliability:** The core offline search engine shall compute correct routing math 100% of the time.
- **Availability:** The V1 offline layover finder shall be available 100% of the time while the extension is active.
- **Maintainability:** The codebase shall remain entirely modular vanilla JavaScript without requiring bundlers.
- **Usability:** The extension shall provide dark mode and glassmorphism styling across all panels.
- **Security:** The extension shall store API keys exclusively in `chrome.storage.local`.
- **Portability:** The extension shall run on any desktop OS supporting Chrome extensions (Manifest V3).
- **Testability:** The deterministic offline search shall be fully testable without external network dependencies.
- **AI Explainability:** The AeroAI Assistant shall clearly indicate to the user when content is AI-generated.

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10. Performance Requirements

- **Maximum response time:** Offline airport autocomplete queries shall complete in < 100ms.
- **Memory usage:** The active extension process shall consume less than 50MB of RAM.
- **CPU utilization:** The flight radar canvas rendering shall consume less than 10% CPU.
- **AI inference latency:** External API calls to the LLM must complete within 5 seconds before a fallback is triggered.

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11. Assumptions

- Users have Google Chrome installed.

- Users can provide valid API keys for the AeroAI Assistant and weather integration (V2).
- The provided offline airport database is sufficiently accurate and up-to-date.
- Stopover policies for supported airlines remain substantially similar to current implementations.

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12. Constraints

- **Programming language:** Vanilla JavaScript, HTML, CSS.
- **Framework:** No external frameworks or bundlers.
- **Extension standard:** Manifest V3 API.
- **Storage:** Data persistence restricted to `chrome.storage.local`.
- **External APIs:** Relies on third-party LLM and weather APIs for V2 functionality.

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13. External Interfaces

User Interfaces

- Chrome Extension Popup (HTML/CSS/JS)
- Interactive Flight Radar Canvas

Hardware Interfaces

- **To Be Completed** (Typically None for a browser extension beyond standard mouse/keyboard input).

Software Interfaces

- Chrome Extension API (Manifest V3, `chrome.storage.local`)

Communication Interfaces

- HTTP/HTTPS for external API requests (AeroAI, Weather) and airline booking redirects.

External Services

- AI LLM API (User-provided via `aiApiKey`)
- Weather API (User-provided via `weatherApiKey`)

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14. Requirements Traceability Matrix

Requirement ID	Level-2 Capability	Requirement Description
FR-1.1.1	1.1 Autocomplete Airport Selection	Autocomplete within 100ms.
FR-1.2.1	1.2 Select Departure Date	Date management within calendar bounds.
FR-1.3.1	1.3 Adjust Stopover Duration	Slider constraints (1-14 days).
FR-2.1.1	2.1 Calculate Haversine Detour	Haversine calculation within 1 second.
FR-2.2.1	2.2 Rank Stopover Hubs	Rank execution within database limits.
FR-3.1.1	3.1 Plot Candidate Flight Paths	Plot paths on canvas.
FR-3.2.1	3.2 Update Active Coordinates	Coordinate updates during radar sweeps.
FR-4.1.1	4.1 Construct Airline Booking URL	URL construction using GET parameters.
FR-5.1.1	5.1 Fetch Visa Requirements	Visa fetch within 5s.
FR-5.2.1	5.2 Fetch Weather Data	Weather fetch within 5s.
FR-5.3.1	5.3 Generate Day-by-Day Itinerary	Itinerary generation within rate limits.
FR-6.1.1	6.1 Detect API Failure	Detection within 5s of API call.
FR-6.2.1	6.2 Trigger Deterministic Fallback	Transition to fallback within 1s.

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15. Future Versions

- **Version 1:** Deterministic layover finder. Airport search, Haversine detour engine, static stopover-program scoring, flight radar, booking link generation. Fully offline.
- **Version 2:** Adds the AeroAI Advisor (AI-assisted layover evaluation) and live trip context (visa, currency, weather). Falls back to V1 ranking when a service is unavailable.
- **Version 3: To Be Completed**
- **Future enhancements: To Be Completed**

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16. Open Issues

- **To Be Completed**: Which specific LLM API and Weather API providers are targeted by default?
- **To Be Completed**: Are there limits on the maximum distance evaluated by the Haversine detour engine?

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17. Glossary

- **Manifest V3:** The latest extension platform specification for Google Chrome.
- **Haversine Formula:** Mathematical formula used to calculate the great-circle distance between two points on a sphere given their longitudes and latitudes.
- **AeroAI Assistant:** The V2 intelligent assistant module that provides contextual information (visa, weather, itinerary) for a layover.
- **Deterministic Ranking:** The V1 offline logic that ranks layovers based on fixed math (detour distance, savings) rather than AI inference.
- **Hidden-city / Skiplagging:** A ticketing strategy where a passenger disembarks at a connection rather than the final destination (explicitly excluded from this project).